

transmitter (Tx) and receiver module.

The same technology can be configured for heterogeneous mode, enabling wireless communication between microcontrollers from different families, such as Arduino to ESP, AVR to PIC, or ESP to STM. Fig. 5 shows an alternative circuit which is an example of heterogeneous mode where you can use the WeMos Mini and Arduino to establish communication between two different

families of MCUs.

The transmitter encodes a user-provided string into ASCII, augments each character with extra bits, and transmits them as modulated laser pulses. The receiver (WeMos D1 Mini Lite) captures light variations, decodes the signals back into ASCII values, and reconstructs the original message.

Designing the system

The design consists of two parts: a transmitter module and a receiver module. While it is possible to integrate both a laser and a photodiode at each end—allowing both devices to function as transmitters and receivers for full-duplex communication—this system is intentionally kept simple. Each device is configured for a specific role: either as a transmitter or a receiver.

Transmitter. The transmitter initialisation begins by setting the laser module (connected to `LED\_PIN`) to a high state, switching the laser on. The laser remains on in idle mode, indicating that the system is ready to transmit. Fig. 6 shows the snippet of the transmitter source code.

To ensure proper input handling, the transmitter reads a string from the serial monitor (until a newline

character `\n`, triggered by pressing Enter). The input string is then displayed on the transmitter's serial monitor for verification.

Each character is converted into its corresponding ASCII code (a byte), with a start bit (0, used as a marker) added at the beginning and a stop bit (1) padded at the end, forming a 10-bit frame (1 start bit + 8 data bits + 1 stop bit).

Transmission begins with the start bit, during which the laser is turned off for the predefined `BITDELAY` duration. Subsequently, each of the 10 bits is transmitted individually. If a bit is 1, the laser is turned on; if a bit is 0, it is turned off—each for a duration of `BITDELAY` milliseconds. Since the stop bit is 1, the laser remains on for an additional `BITDELAY` period, marking the end of the transmission frame.

Receiver. The receiver starts by monitoring the state of the light received through the integrated photodiode. To ensure correct interpretation, two variables—representing the previous and current signal states (indicating laser ON)—are both initialised to 1. Synchronisation with the transmitter is achieved by detecting a transition in the optical signal, specifically a shift from laser ON (logic 1) to laser OFF (logic 0). This transition marks the start of a transmission.

To facilitate positive logic interpretation, an additional function is incorporated that inverts the incoming signal. Thus, a transition from laser ON (logic 0) to laser OFF (logic 1) is used to represent signal changes in positive logic terms. Fig. 7 shows the snippet of the receiver source code.

Synchronisation and data reconstruction. Next, the receiver begins reading the incoming character. A deliberate delay of 1.5 times the BITDELAY is introduced before reading the following bits.

```

1  #define PHOTO_DIODE_PIN D5
2  #define BITDELAY 3
3  int previous_state = 1;
4  int current_state = 1;
5  void setup() {
6  Serial.begin(9600);
7  }
8
9  void loop() {
10 current_state = get_bit();
11 if (!current_state && previous_state) {
12 Serial.print(get_byte());
13 }
14 previous_state = current_state;
15 }
16 int get_bit() {
17 int val = digitalRead(PHOTO_DIODE_PIN);
18 return 1-val;
19 }
20 char get_byte() {
21 char byte = 0;
22 delay(BITDELAY * 1.5);
23 for (int i = 0; i < 8; i++) {
24 byte = byte | (get_bit() << i);
25 delay(BITDELAY);
26 }
27 return byte;
28 }

```

Fig. 7: Snippet of the receiver source code

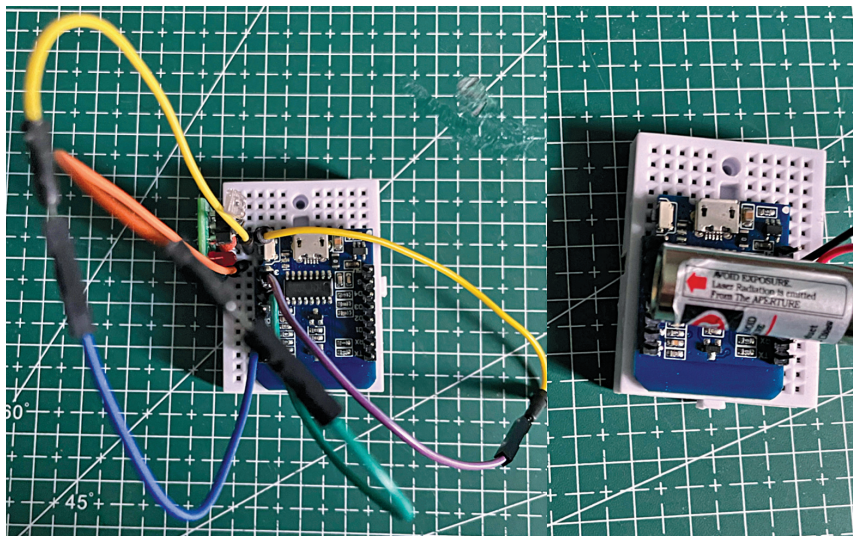


Fig. 8: EFY lab's prototype, left is the receiver and right is the transmitter